

Homework Activity

Scenario 1: you are a Network Planner for Emergency Communication

Situation:

A major natural disaster has occurred in a city. Several electricity lines and communication towers have been damaged. One of the major cellular towers is completely unavailable, while thousands of people are trying to make emergency calls and access mobile internet.

Your Role

You are the Emergency Communication Network Planner responsible for restoring mobile connectivity and ensuring that emergency calls and access mobile internet.

Tasks

1. Identify the problems caused by the failure of a cellular tower

- Sol: Users within the failed BTS's coverage area may lose cellular connectivity
- Ongoing voice calls may be dropped
 - Mobile internet services may become unavailable or extremely slow
 - Emergency services may have difficulty communicating with hospitals, police, fire

departments and
• Neighbouring
displaced us

2. Role of the

- Identify the area
- Estimate
- check ne
- Redirect
- plan ten

3. Assumpti

Parameters

Disaster
population

Failed

Neighbour

Normal

Emergency

departments and rescue teams.

- Neighboring BTSs may become overlapped because displaced users simultaneously attempt calls.

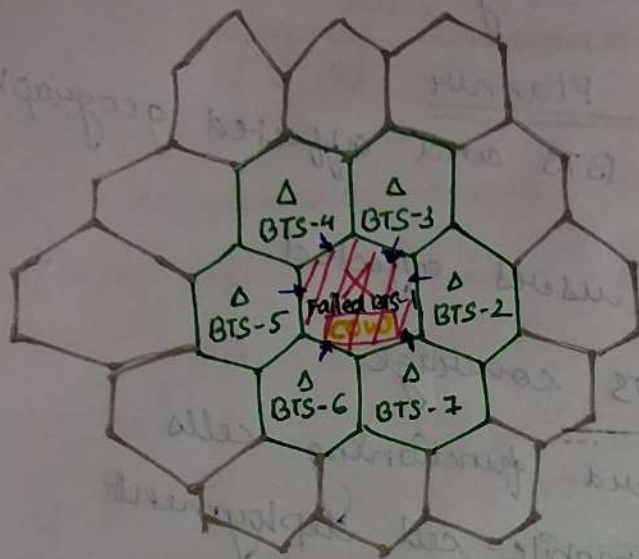
2. Role of the Network Planner

- Identify the failed BTS and affected geographical area
- Estimate the no. of users affected
- check neighbouring BTS coverage
- Redirect users toward functioning cells
- plan temporary BTS/mobile-cell deployment

3. Assumptions

Parameter	Assumption
Disaster area	5 km x 5 km
Population affected	50,000 people
Failed BTS	1
Neighbouring BTSs	3
Normal cell radius	Approximately 1-2 km
Emergency traffic	very high

4. proposed cellular Network diagram



Core: MSC / Gateway + Backup
power (restoration &
priority control)

BSC (controls BTS-2 - BTS-7)
re-parks users, traffic
rerouted

Emergency services: Police,
Fire, hospital Authority
access

Legend

- Failed cell (no service)
- working neighbour cell
- outer cells (unaffected)
- cow = cell on - wheels temporary BTS
- Neighbours extend coverage (power-up, handoff / cell
reselection)

5. cell selection and placement

The normal layout is a hexagonal 7-cell cluster.
When BTS-1 fails its cell becomes a coverage hole
~~some~~ surrounded by six healthy cells.

- Increase transmit power of the 6 sectors facing the hole
(for example +3 to +6 dB, within safe limits).
- up - tilt / re-orient antennas of BTS-2 to BTS-7 toward
the failed area, so their cell radius grows and
overlaps the hole ('cell breathing')

6. BTS Placement

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buildings

- connect
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fibre

7. Frequency

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8. Handover

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6. BTS Placement

- place the cow at the old site if the tower area, is safe, otherwise on open high ground close to the centre of the hole, away from damaged buildings and flooding
- connect it using microwave or satellite backhaul to a healthy neighbouring BTS (BTS-3 or BTS-5), because fibre may be cut.

7. Frequency Reuse strategy

- Neighbouring cells keep their own frequency sets (F_2 to F_7) so that no new interference is created.
- The cow takes the frequency set F_1 of the failed BTS-1 so the reuse pattern of the 7-cell cluster stays unchanged and co-channel interference to other clusters is avoided.
- while neighbours are power-boosted, reuse distance is monitored; if interference rises, power is reduced again as soon as the cow is on air.

8. Handoff / Mobility Analysis

- Users who were in an active call on BTS-1 cannot be handed off, because the serving BTS is dead; the call drops and the phone re-selects the strongest neighbour in idle mode.
- Handoff parameters (threshold, hysteresis, time-to-trigger) are tuned to avoid ping-pong between overloaded cells.

9. Coverage and Capacity Analysis

Coverage: boosting neighbouring restores roughly 60 to 70% of the hole (assumption) and the CoW restores the rest

Item	Normal	Disaster
Load per neighbour cell	30 Erlang (60%)	105-175 Erlang
coverage of hole	100% by BTS-1	0% (only edge)
call blocking	Below 2%	every high

10. Problems / challenges identified

- Loss of power and backhaul; fuel shortage for generators
- Severe congestion and call blocking; mobile data and voice compete
- Interference caused by over-boosted neighbours.

11. proposed solutions

- Traffic surge: use SMS and short text instead of voice, restrict video / data, reduce call duration, use half-rate speech coders, Access class Barring for normal users and cell-broadcast alerts
- Redundancy: overlapping cells, dual-homed backhaul / fibre + microwave links, N+1 power with batteries
- Alternative emergency communication: TETRA / walkie-talkie radios.

Conclusion

A failed BTS creates a coverage hole and overloads its neighbours. By tuning neighbouring cells, deploying

a CoW, using emergency

Scenario 2: S

1. problem I

A growing for IoT devices public man The challenge capacity services

2. Role of

- Divide
- Identify
- plan and

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a cow, using redundancy and giving priority to emergency traffic.

Scenario 2: Smart City Mobile Network Planner

1. Problem Identification

A growing smart city needs reliable mobile connectivity for ~~ten~~ millions of phones, and a very large no. of IOT devices (traffic signals, CCTV, smart parking, sensors, public transport, hospitals and educational institutions). The challenge is to give wide coverage, very high capacity in crowded places, low delay for critical services and low interference.

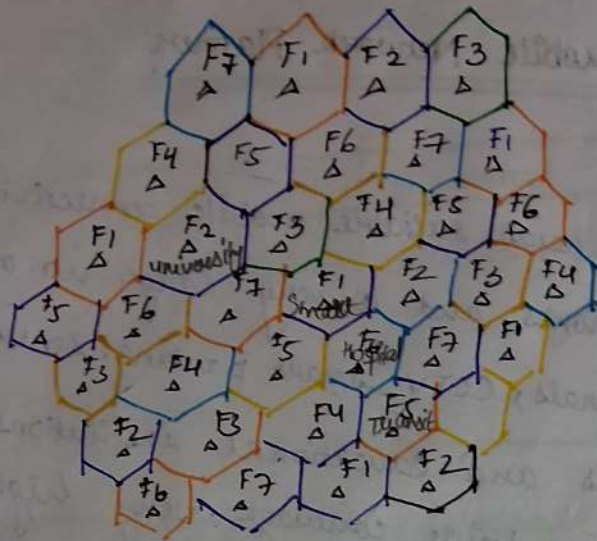
2. Role of the Network Planner

- Divide the city into cells and decide cell sizes
- Identify hotspots and decide BTS and small-cell sites
- Plan frequency reuse and interference control, mobility and handoff

3. Assumptions

- city area about 150 sq km with about 2 million users and 100,000+ IOT devices.
- Zones: CBD / commercial, residential, industrial / IT park
- 4G/5G network with macro, micro, pico / small cells; fibre backhaul is available in the city
- 7-cell frequency reuse for macro layer

4. proposed cellular Network Diagram



Smart-city Core

Fibre / Microwave / Medium

Macro
BTS

Micro
cells

Mobile
users

IoT devices

Smart junction
density zones

5. cell selection and placement

Area	Cell type/size	Reason
Suburbs, residential, Industrial	Macro cell, 2-3 Km radius	Moderate traffic, wide coverage at lower cost
Dense CBD, malls, offices	Micro cells, 200-500 m	High traffic, needs frequency reuse in small areas
Stadium, Station hospital, crowded junctions	Pico/Small cell, 10- 100 m	very high density, indoor coverage
Highway / ring roads	Elongated macro sectors along road	fewer handoffs for fast vehicles

6. BTS placement

- Macro BTS on tall rooftops or towers at the centre of each hexagon, 3-sector antennas.
- Micro cells on street poles / building walls at about 5 to 10 m height in the CBD
- small cell inside malls, stations, hospitals and on smart light poles near crowded junctions, linked by fibre.

7. Frequency reuse strategy

- Macro layer uses a 7-cell cluster (F_1 to F_7); cells that share a frequency are separated by the reuse distance $D = R \sqrt{3N}$, which is about $4.6R$ for $N=7$
- small cells use a different carrier band
- sectorisation (3 sectors) increases capacity and reduces co-channel interference

8. Handoff / Mobility Analysis

- pedestrians: slow movement, handoff between micro / small cells based on signal strength and hysteresis
- Interferences sources and handoff failures are monitored by SON (self-organising network algorithms)
- Layer handoff: a vehicle that slows down (traffic jam) can move to micro cells (load-based handoff)

9. Coverage and Capacity analysis

Coverage: macro cells provide continuous coverage with about 10 to 15% overlap.

Capacity: demand per sq km in the CBD can be 50 to 100 times that of residential areas.

10. Problems / challenges identified

- ~~the~~ co-channel Interference between cells using the same frequency; adjacent-channel Interference
- cross-tier interference between macro and small cells; interference from tall buildings
- Congestion in crowded places and at events; frequent handoffs for vehicles

11. proposed solutions

- Interference: proper reuse distance, sector antennas with down-tilt, power control, ICIC (eICIC), and separate carriers for tiers

Reliability: redundant fibre ring backhaul, battery backup and a monitored core network

Dense areas: small cells on poles and inside buildings in CBD, stations and stadiums.

12. Conclusion

A layered macro and frequency network is needed for TDD.

Scenario: 3

1. problem

A campus needs outdoor hours for seminar is poor

2. Role

- Idle Capacity
- Dec
- Plan

12. Conclusion

A layered cellular design (macro for coverage, micro and small cells for capacity) with frequency reuse, interference management and network slicing supports both mobile users and IoT devices.

Scenario: 3 University Campus Network Planner

1. Problem Identification

A campus with 20,000 students and 1500 faculty needs reliable coverage in all buildings and outdoor areas. Traffic is moderate during normal hours but jumps during examinations, festivals, seminars and placement drives, and indoor signal is poor in thick-walled buildings.

2. Role of the Network planner

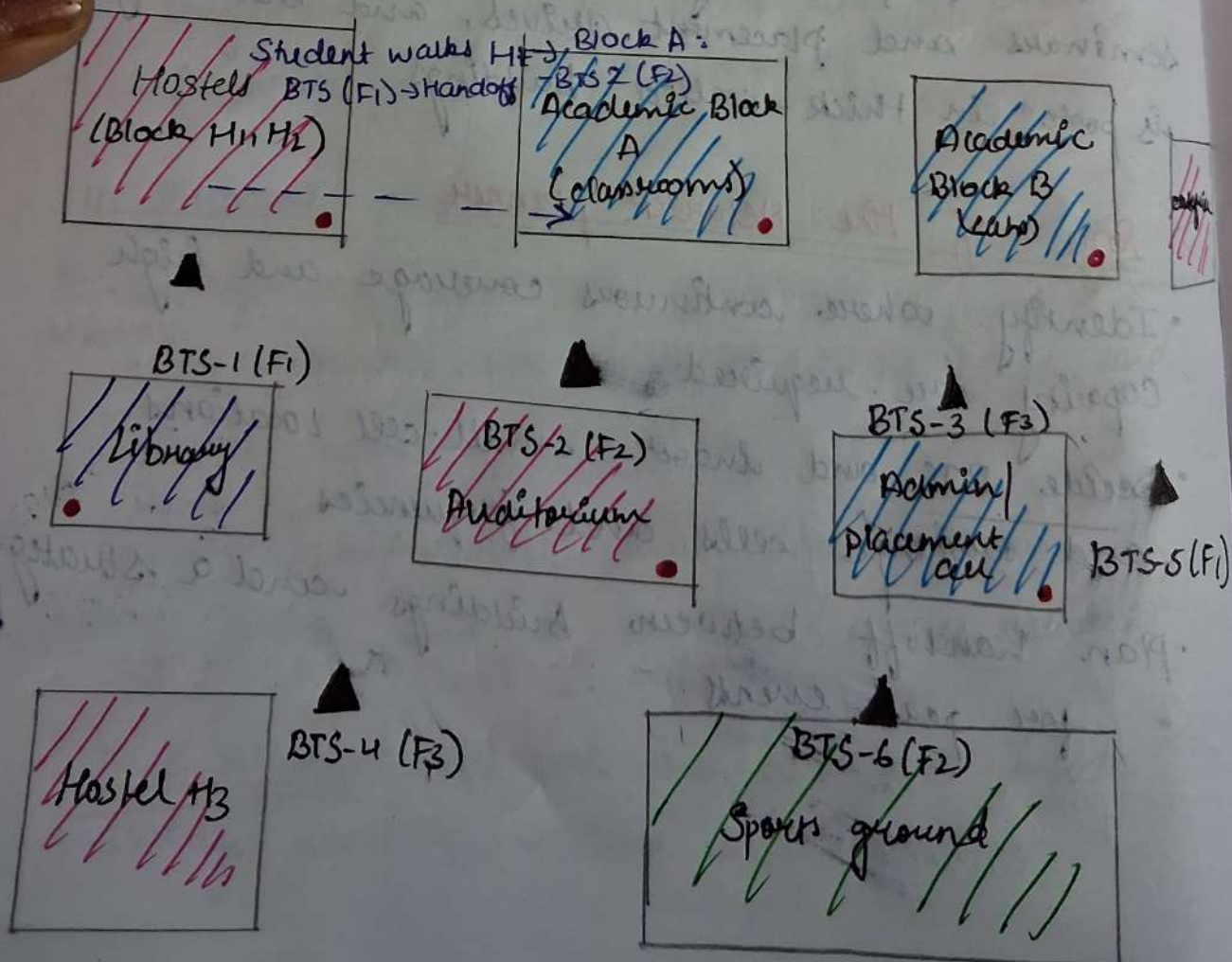
- Identify where continuous coverage and high capacity are required
- Decide BTS and indoor - small-cell locations and arrange cells and frequencies
- Plan handoff between buildings and a strategy for peak events.

3. Assumptions

- Campus about 1.2 Km x 0.75 Km with academic blocks, library, hostels, cafeteria, auditorium, Sports ground and placement block
- 3-cell frequency reuse (F_1, F_2, F_3), cell radius about 200 to 250 m, 10 to 15% overlap
- Operator provides macro/micro sites; fibre backbone inside campus

4. Proposed cellular Network Diagram

○ F_1 cell ○ F_2 cell ○ F_3 cell • Indoor small cell/DAS ▲ macro/micro BTS



5. cell selection and placement

- Continuous coverage required: classrooms, labs, library, hostels, admin/placement block, cafeteria, auditorium corridors and roads and sports ground
- The campus is divided into six overlapping cells (one per BTS) arranged so that each cell covers a group of buildings and adjacent cells overlap slightly for handoff.

6. BTS placement

- BTS-1 near hostels H1/H2; BTS-2 near Academic Block A/library centre; BTS-3 near Academic Block B (labs); BTS-4 near hostel H3/library side; BTS-5 near cafeteria/least edge; BTS-6 near admin and sports ground
- Antennas on rooftops or 15-25 m poles, 3-sector or omni depending on the site
- A spare site (cow) near the auditorium/sports ground for events.

7. Frequency Reuse strategy

- 3-cell reuse: adjacent BTSs use F_1, F_2, F_3 in rotation so adjacent cells never share a frequency
- Cells using the same frequency (e.g. BTS-1 and BTS-5) are far apart, so co-channel interference stays low.

8. Handoff / Mobility Analysis

- The call or data session continues without breaking (soft) seamless handoff in 4G/5G.
- Role of handoff: It maintains continuity of calls, data while users move, balances load between cells, and keeps signal quality above the minimum.
- Moving from outdoor to indoor: handoff between the outdoor macro cell and the indoor small cell ~~with~~ inside the building.

9. Coverage and Capacity Analysis

Six BTSs with about 220m radius cover the campus with overlap. With 3-sector sites there are about 18 sectors; 3200 busy hour users means roughly 100 active users per sector.

10. problems / challenges identified

- Large crowds in the auditorium or ground cause congestion.
- Interference near the edges between cells; handoff failure in gaps between buildings.
- Hostel traffic peaks at night; exams periods increase load.

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11. proposed Solutions

- Improve indoor coverage: indoor small cells, distributed antenna system (DAS), femtocells, repeaters / boosters, Wi-Fi calling and antennas in lifts and corridors.
- Handoff: proper neighbour lists and 10-15% overlap; time hysteresis and thresholds
- Interference: reuse of F1-F3, tilting antennas down power control

12. Conclusion

A six-site, 3-frequency design with indoor small cells gives continuous coverage across the campus. Handoff keeps sessions alive while students move and temporary extra capacity handles examinations, festivals and placement drives.

Scenario 4: Railway Mobile Network Planner

1. Problem Identification

Passengers must get continuous voice, video, payment and internet over a 200 km railway route that passes cities, villages, tunnels, bridges and remote forest/hill areas at high train speed.

2. Role of the Network Planner

- select tower locations along the track and arrange cells in a linear chain
- plan frequency reuse and handoff for high speed users
- solve coverage problems in tunnels, bridges and remote areas, and ensure reliable services.

3. Assumptions

- Train speed 100 to 120 km/h about (33 mph); single railway line with towers along the track
- spacing: city 2-3 km, villages 4-6 km, remote areas 7-8 km with tall towers; about 80 to 60 towers.
- carrier frequency about 2.1 GHz; 3-frequency reuse in a linear arrangement.

A. proposed cellular Network Diagram

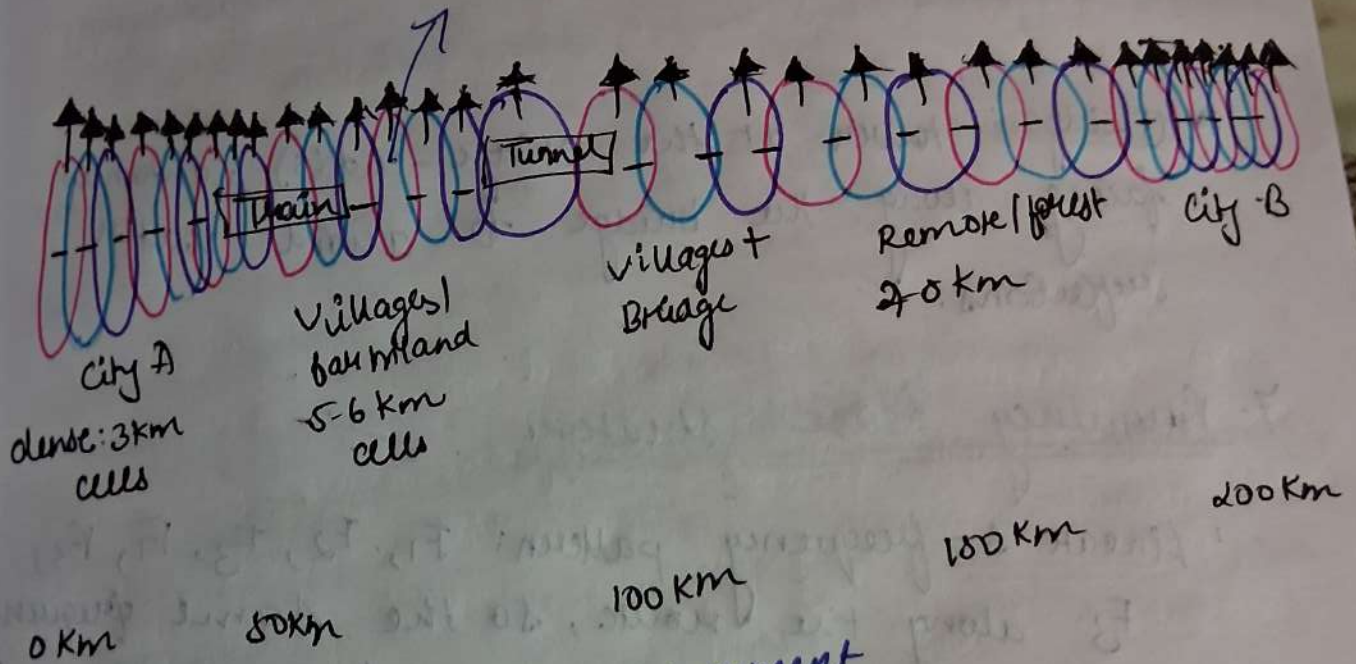
■ F₁ cell (tower below) ■ F₂ cell (tower below) ■ F₃ cell (tower below)

High capacity:
macro + small
cells at
station

Handoff Zone (overlap
+ 5%) n-th cell → (n+1)th
cell, pre-emptive, neighbour
list based with approx

Extended: tall
towers, high gain
antennas

city B/
terminal
station
capacity
cells



5. cell selection and placement

• cells are elongated (oval) along the track, because
waves move only along one line; antennas are
directional, pointing along the railway in both
directions

• cities and stations: small cells (2-3 km) for high
capacity. villages / farmland: medium cells
(4-6 km). Remote areas: large cells (7-8 km)

• Adjacent cells overlap by about 10 to 15%. So
that handoff can complete before the
train leaves the cell.

6. BTS placement

- Towers near the track side (within 100-300m) on high-ground, so that line of sight is available stations get macro BTS plus small ones
- Tunnel: BTS / repeater at both portals plus leaky coaxial cable or distributed antennas inside the tunnel

- 
- Bridge: tower at the bridge end(s), antennas facing along the bridge to avoid water reflections.

7. Frequency Reuse Strategy

- Linear 3 frequency pattern: $F_1, F_2, F_3, F_1, F_2, F_3$ along the route, so the same frequency repeats after every three cells (reuse distance about 3 cells lengths)

- Adjacent cells never share a frequency, so interference at the overlap region is minimised each frequency group is reused many times over 200 km.

8. Handoff / Mobility Analysis

At 120 km/h and 3 km cell length, the train crosses a cell in about 90 seconds, so handoff occurs every 1.5 to 3 minutes

- High speed needs: early / predictive handoff based on the known track direction, make before break (soft handoff)

9. Coverage and Capacity Analysis

Coverage: continuous strip coverage along the track, with a signal margin to cover carriage penetration loss (15 to 25 dB).

Capacity: a full train (about 1,000 passengers) all using mobile data produces a sudden load on one cell at a time; stations and cities need capacity cells, while remote areas need only basic coverage.

10. Problems / challenges Identified

Calls drops: late or failed handoff at high speed coverage holes in cuttings / tunnels / forests, congestion at stations, Doppler shift.

- Tunnels and obstacles: tunnel walls and hills block the radio signals completely (no line of sight) bridges and cuttings cause reflections and shadowing

11. Proposed Solutions.

- Tunnels: leaky feeder cable or DAS inside, repeaters / small cells at tunnel entrances so that handoff happens before entering

- Remote areas: taller towers, high-gain directional antennas, remote radio heads.

- High-speed reliability: train mounted mobile relay (onboard wi-fi)

- Interference: 3-frequency reuse, directional antennas pointed along the track, power control

- Redundancy: dual backhaul and battery backup at every tower

12. Conclusion:

A linear chain of elongated cells with 3-frequency reuse, proper overlap, predictive handoff, tunnel DAS and strong remote-area design gives continuous connectivity along the 200km route.

Scenario: 5 Indoor Cellular Network Planner for a Shopping Mall

1. Problem Identification

The mall has about 30,000 visitors per day (more on weekends and festivals) but users complain about weak signals and slow internet. The outdoor network cannot penetrate the building well and the indoor capacity is not enough for payments, video, social media and navigation.

2. Role of the Network Planner

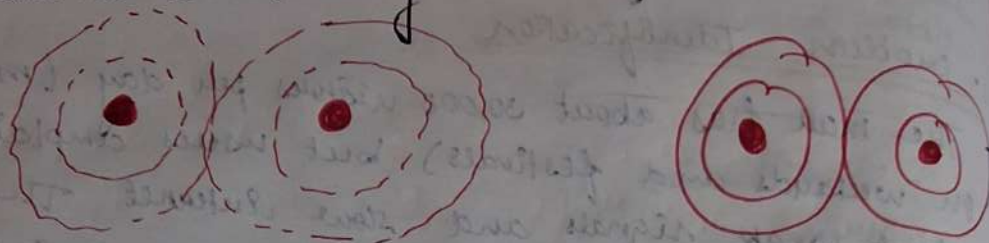
- Find the reasons for weak signal and congestion; decide whether coverage and capacity are required.
- Design an indoor network with small cells / DAS, plan interference control and handling of traffic peaks.
- Ensure seamless mobility between floors and from outdoors to indoors.

3. Assumptions

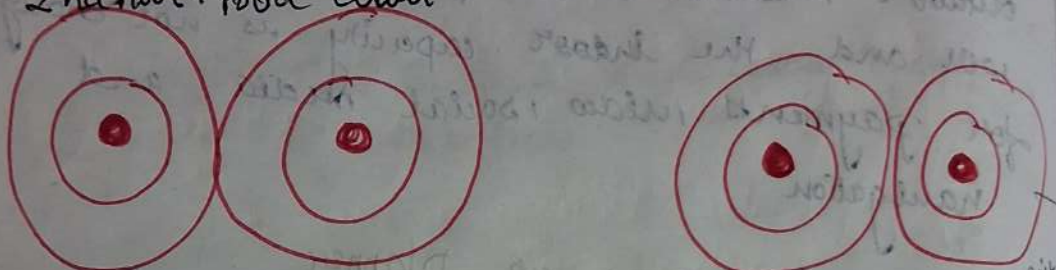
- Mall with basement parking and 4 floors and a central atrium.
- Peak weekend load about 4,000 to 5,000 simultaneous users; heavy mobile data use.
- Operators share the infrastructure (central host) with fibre backhaul and a BBU room in the basement.

4- proposed cellular Network

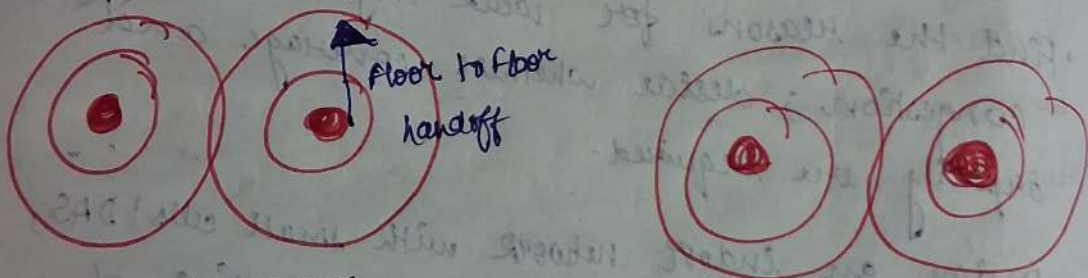
3rd Floor: cinema / cramming



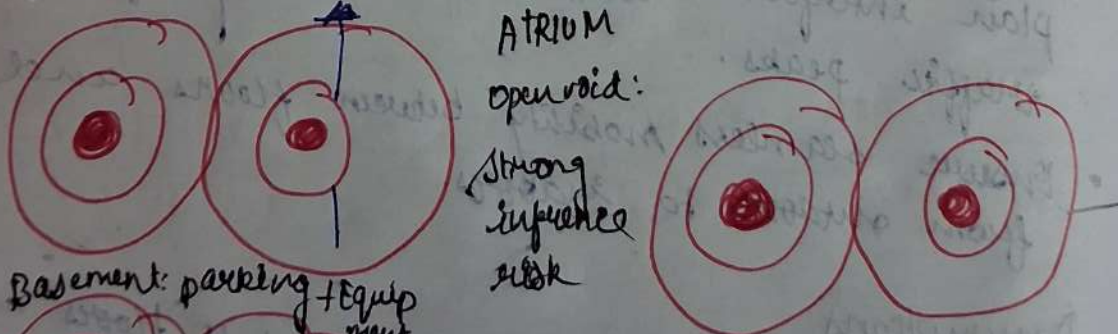
2nd Floor: food court



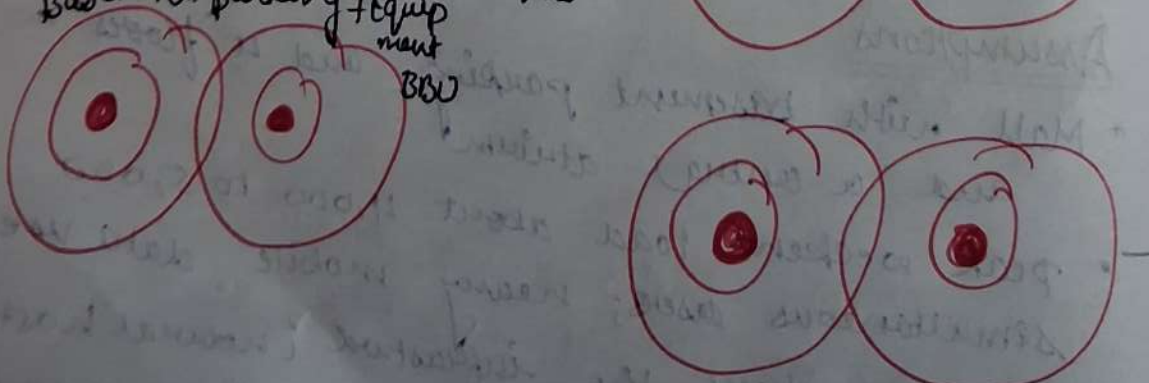
1st Floor: Fashion, Electronics



Ground Floor: Entrance



Basement: parking + Equipment
BBU



BBU hub
+ backhaul
to core

Red dot = Base station
Circled sign =

5. cell selection

Material

concrete / reinforced wall

Metal-coated low-e glass

Left shaft / metal closure

Human crowd

6. BTS

small corridor
even cinema

EXT

BB

in

me

- Red dot = Indoor small cell / DAS antenna
- Clotted ring = coverage radius separate PC / frequency reuse between floors

5. cell selection and placement

Material	Typical loss	Effect
concrete / reinforced wall	10-25 dB	Strong attenuation, blocks inner areas
Metal-coated / low-e glass	20-40 dB	Almost blocks signals
Lift shaft / metal doors	20-30 dB	Dead zones
Human crowd	5-5 dB	Extra loss at peak time

6. BTS placement

- Small cells or DAS antennas on ceilings along corridors on every floor, one cluster in every anchor store, the food court and cinema
- Extra antennas in lift shaft, parking and staircase
- BBU / baseband hub in the basement equipment room connected by fibre risers to the remote radio unit; an outdoor macrocell

BBU hub
backhaul
to core

done on the roof before connection to the operator network

7. Frequency Reuse strategy

- A dedicated indoor carrier is used so that the indoor and outdoor networks do not interfere
- Low power small cells (about 20 to 250 W) keep the interference radius small
- Each floor uses a different cell identity pattern from the floor above and below and the same frequency is repeated only where there is enough wall/floor isolation.

8. Handoff / Mobility Analysis

- At the mall entrance, a handoff takes place between the outdoor macro cell and the indoor cell.
- Antennas at stairs and lift lobbies give an overlap area so handoff finishes before the signal falls.
- A user moving between floors by escalator/staircase or lift passes from the

Small cells

9. Coverage

Aspect	
Coveral	
Area	
Solution	
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Small cell on one floor to the next.

9. Coverage and Capacity

Aspect	Coverage requirement	Capacity requirement
Goal	Signal everywhere	Enough throughput for many users at once
Area	corridors, stores, lifts, parking	Food court.
Solution	Well-placed antennas/DAS repeaters	Small cells, carrier reuse
peak need	constant	weekends and festivals

10. Problems / challenges Identified

- Interference: outdoor macro signal leaking through glass latrines, overlap between floors, overlapping small cells
- Shadowing in lifts, basements and parking; difficult testing, high installation cost and multi-operator coordination

11. proposed solution

- Small cells improve indoor connectivity because they place ~~the~~ place antenna close to the user avoid wall penetration loss

giving higher frequencies, ~~and~~ capacity by
reusing frequency

- Multoperator: neutral-host DAS shared
by all operators

- Interference: careful PCI/frequency
planning, power control, a dedicated indoor
carrier and placement.

12. Conclusion

An indoor network small of cells / ~~DAS~~ DAS

with proper frequency planning, floor to floor

handoff and extra capacity at hotspots

fixes the weak signal and

slow internet problem of mall